

Chromatography

Method allow us to sort proteins by size or by how they cling (bind) or separate from other substances.

Long glass tubes are filled with microscopic resin beads and buffered solutions.

The protein extract is then added and flow through the resin beads in a glass column.

Depending on the resin used, the protein either sticks to beads or passes through the column while the beads act as Filtration system.

Size – exclusion Chromatography (SEC)

Uses gel bead as Filtration system, larger protein molecules Quickly work their way around the gel beads while smaller molecules pass through more slowly, because they are able to slip through tiny holes in the beads.

The gel are available in a variety of pore sizes, the necessary gel for proper separation depend on the molecular weight of the contaminants or protein being separated.

This method can make only preliminary separations.

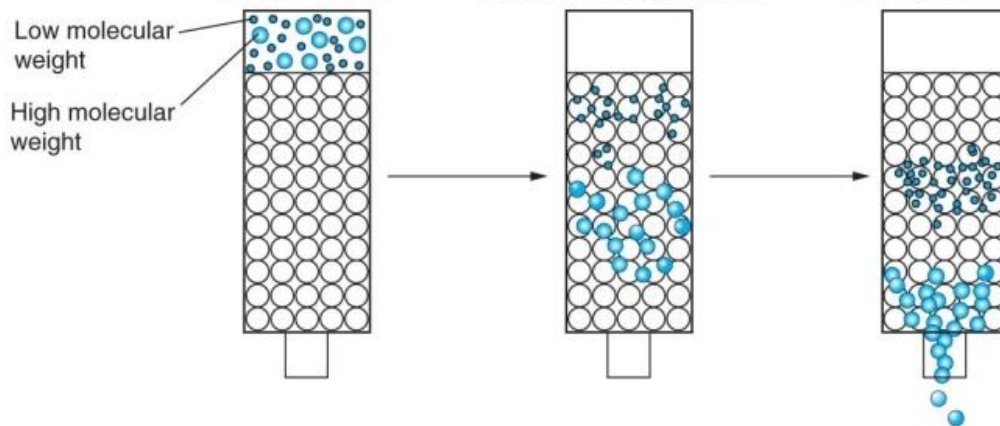
Require very large columns (problem).

(a)

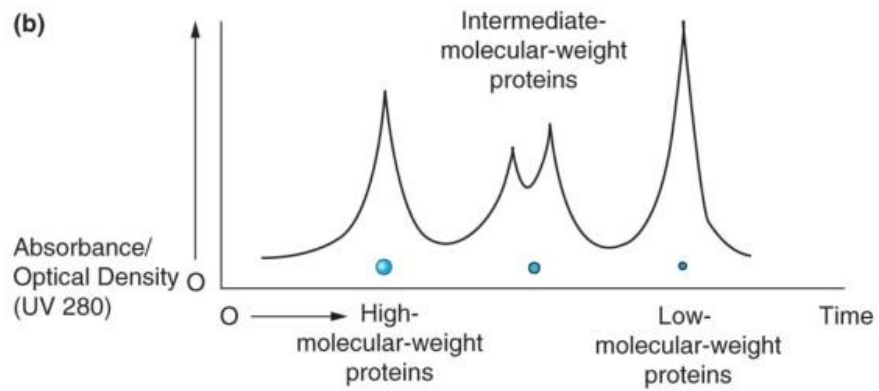
Sample of low-molecular-weight and high-molecular-weight proteins applied to top of column

As buffer flows, the low-molecular-weight proteins enter the gel beads, retarding their flow

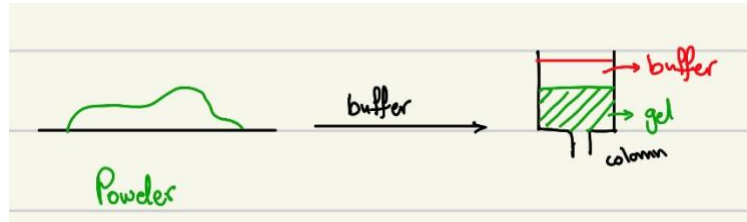
The higher molecular weight proteins are less able to enter the gel beads, so they elute first



(b)



The column chromatography is prepared from powder called **resin**, so we add buffer to the resin by a process called swelling and it becomes gel (this gel is colorless) then we add this gel to the column



We have something called equilibration and it's the process of adding more buffer to allow the buffer to go through the column several times to make sure that buffer will cover the inside of the column

The next step would be to add the protein mixture into the column

- the protein mixture consists of many protein (**your protein and other protein**)

this step is called the **loading step**

the loading mean: adding the protein mixture to the top of column, and the protein will start moving from the top to the bottom, why is that?

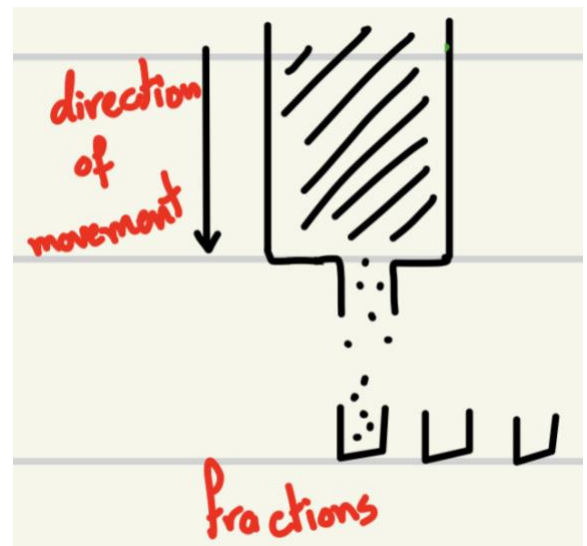
This direction of movement allows some protein to stuck to the resin and some will go through

After we do that, we come to a new step called **elution**

Elution is the process of extracting the protein from the column

Now the main goal of this process will be as we said to extract the protein from column by fraction, our big hope is to get our protein inside the same fraction

In the end all the protein will be extracted from the column, during this process we add liquid to push the protein outside the column.



You have to know that the liquid is colorless, and the protein is colorless, also the column is colorless so when we do the elution, we are extracting colorless liquid in small container (fraction) and they will have small volume

Now we are going to talk about a new type of chromatography: **ion exchange chromatography**

- Ion exchange chromatography- relies on electrostatic charge to bind proteins to resin beads in the column
 - When charged proteins bind to resin then the contaminants pass through and out of the column
 - Proteins can then be released from column by changing electrostatic charge by increasing salt concentrations of salt solutions
 - » Anion exchange resin is positively charged
 - » Cation exchange resin is negatively charged

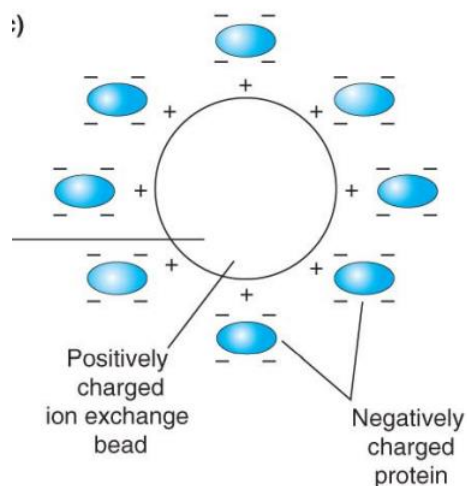
This technique will separate the protein based on the electrostatic charge, some protein will have a positive net charge, other will have negative net charge, and other will be neutral

Now why some proteins have positive charge, and some have negative charge?

It depends on the side chain of the protein

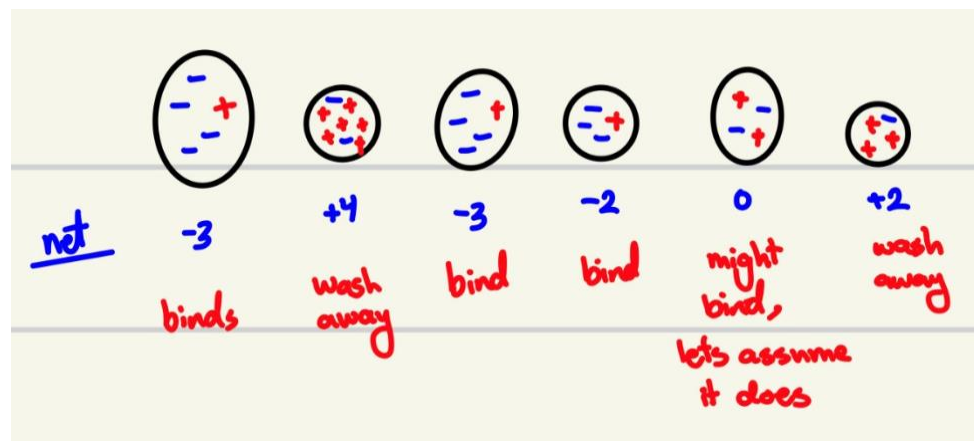
We said net charge because there are another charges on the protein if the total amount of positive charge is higher than the negative charge then the total will be positive

Now not all the protein with the same charge are the same because even if they have the same charge, they could have different magnitude of this charge



In this technique we will use something called microscopic beads which consist of resin, and these beads will have charge on its surface it could be positive or negative, in this example the beads will have positive charge

So, let's assume that we have six proteins:



If we load these proteins on the top what will happen is that some protein will bind to the resin and other will not bind, remember we said in this example the resin will be positive so the proteins that will bind will be the negative one

Wash out means that after we add the proteins, we will add buffer this buffer will wash the positive proteins (let the protein to continue moving), and the protein with negative charge they will be stuck with the resin

Now we will have three proteins and these proteins will be binding to the resin now what we need to do is to elute them from the column

The next step would be to add another type of buffer that will have high concentrations of salt, let's say that the buffer has NaCl because the NaCl is in buffer we will have Na⁺, Cl⁻ ions now the Cl⁻ ions have a negative charge it loves to bind to the resin which means that the Cl⁻ will come and push the protein that has a negative charge away from the resin and this is why this technique is called ion exchange

Now what will happen after the addition of the buffer is that the negatively charged protein detaches from the resin and comes out but not all the protein will leave at the same time

Note:

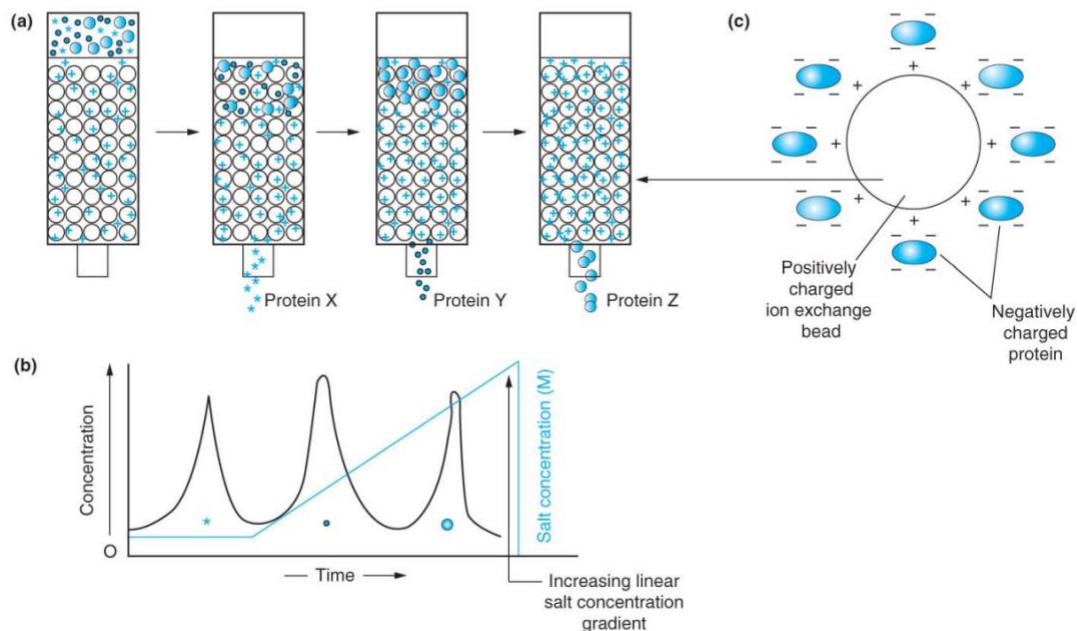
The positively charged protein will be washed out at the same time

You have to know that the elution buffer doesn't have a fixed concentration of salt which means that we can change the concentration as we want this will allow the leaving of the protein gradually (not at the same time)

Now we will start with low concentration of salt the first protein to leave or detach from the resin would be the one that is forming the weakest bond (the lowest negativity) in this case it would be the protein with 0 charge, after that we will start increasing the concentration of salt the next protein to detach would be the -2 protein and then the last protein that will detach is the -3 protein

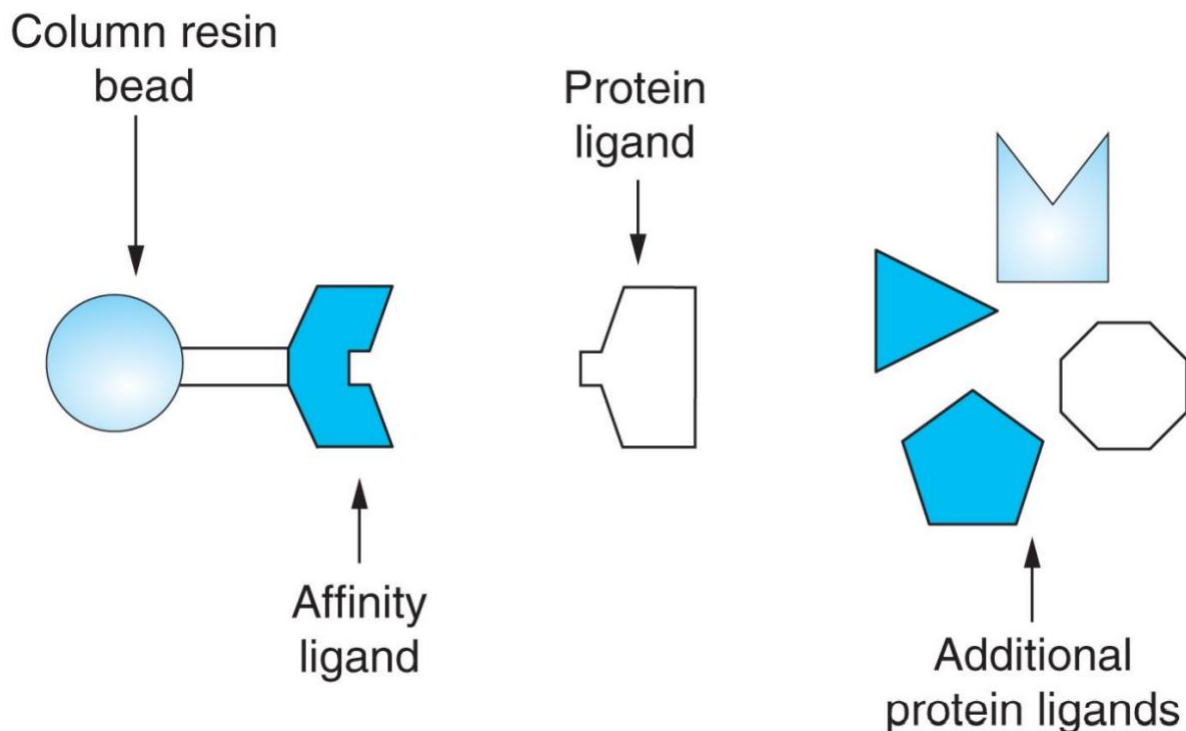
Note:

The same protein could have different charge this would be because of the pH buffer that the protein that the protein exist in



- **Affinity chromatography**- relies on ability of most proteins to bind specifically and reversibly to ligands- small molecules that bind to a particular large molecule in a protein (lock: key model)
 - After proteins have bound to resin, buffer solution is used to wash out unbound molecules
 - Then special buffers are used to break ligand bonds of the retained protein
- » Quite often used for purification of fusion proteins
- * this technique shortens purification process by reducing number of steps

This type of chromatography depends on the affinity between chemical group of the resin and the protein that you want to separate



There will be ligand that will bind only to your protein so its like a tag that you can add it to your protein and the bead will have affinity ligand so only your protein will bind to the bead and the other protein will not, for example we can add his tag

His tag → when you express your protein you add the his (you can add it at the start of the protein or at the end), and this his consist of 6 codon of histidine that normally shouldn't be in your protein but we use them as a tag that will allow the protein to bind to the column

So, for example if we have nickel column the resin will have nickel ions in the surface and these nickel ions can bind to the histidine side chain imidazole, so any protein with histidine can bind

By adding a tag, you can make the purification easier

Now the ligand could be any ligand, or they could be substrate of the enzyme if your protein is an enzyme, or they could be anti-bodies that can specifically catch and bind to the protein

Hydrophobic interaction chromatography

- Proteins are sorted on basis of their repulsion of water

- » Column beads are coated with hydrophobic molecules and the hydrophobic amino acids in a protein are attracted to similar chemicals in the beads

Most of the protein are amphipathic, which means they have parts that are hydrophobic and other parts are hydrophilic

Most of the protein are surrounded by aqueous solution (for example water)

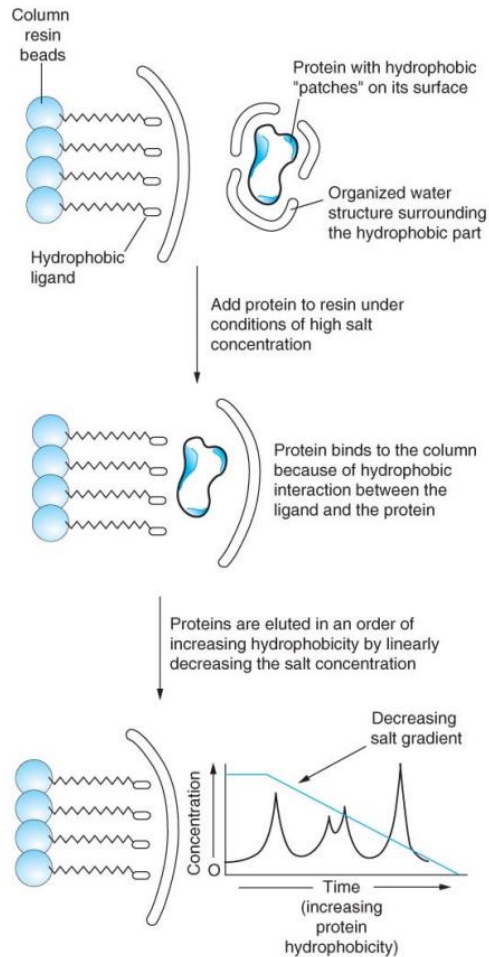
The majority of the hydrophobic parts will be inside the protein, but some hydrophobic amino acids can be found on the surface, and they will be close to each other because they don't love water so they will form hydrophobic patches

In the hydrophobic interaction chromatography, the resin will contain hydrophobic molecules so they can bind to the hydrophobic patches on protein so the protein with higher hydrophobic patches will bind tightly to the resin that contain hydrophobic

molecules, so the protein with low number of hydrophobic patches will be washed away

The first step would be to load the protein on the top of column and as we said that the protein with high number of hydrophobic patches will bind to the bead that contain hydrophobic molecules after that we will use a buffer with high concentration salt to make sure that protein will bind to the beads

The next step will be to lower the salt concentrations because if we lower the salt concentration more water molecules will cover the hydrophobic patches (they will cover only without binding) so that the protein with the lowest hydrophobic patches will be removed first after that we will lower the concentration more so the protein with the higher hydrophobic patches will start to be removed one by one.



Isoelectric focusing

a Technique separate the proteins based on Isoelectric point.

Isoelectric point (pI /IEP) where charge on protein match the PH of the solution.

Each protein has a specific number of charged amino acids on its surface in specific place, because of this unique combination of charged groups, each protein has **Isoelectric point**.

The IEP can be used to separate similar proteins from one other

Two – dimensional electrophoresis separate proteins based on their electrical charge and size.

Procedure steps :

Put proteins on the gel or filter paper, gel or paper has a gradient for one side of paper or gel low PH, for other side high pH

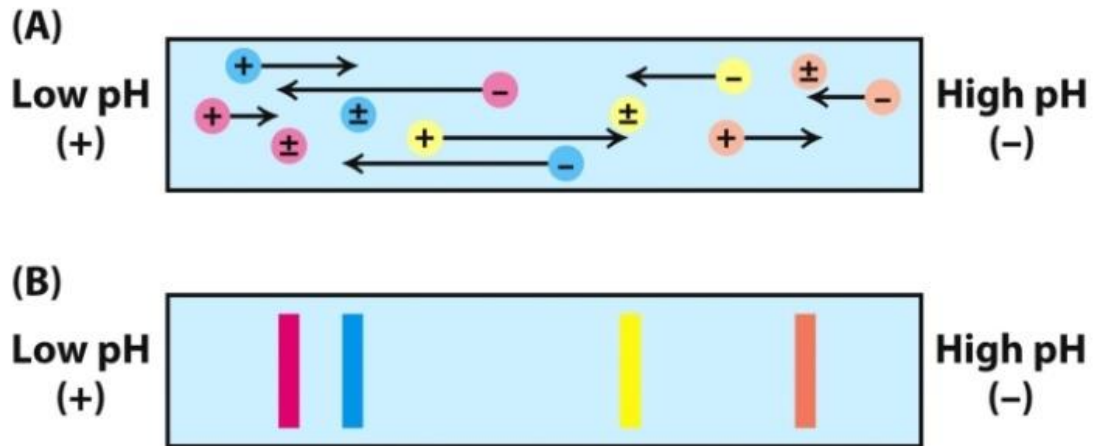
If we have mixture of proteins that have different PI

If we would separate these proteins from each other, we will put them in the middle of gel (PH= 7.5), if the PI for each protein is smaller than PH, protein will have a negative charge, it will start move to ward low PH side, low PH = positive charge.

If the pi of protein is higher than pH (we put protein at the middle PH=7.5), the protein will have positive charge, bit start move to ward high PH, High PH = negative Charge.

proteins start move until the PI match the PH, if this happened the protein will stop.

Look to the below figure



High – performance liquid Chromatography
HPLC)

Analytical method used for proteins separation

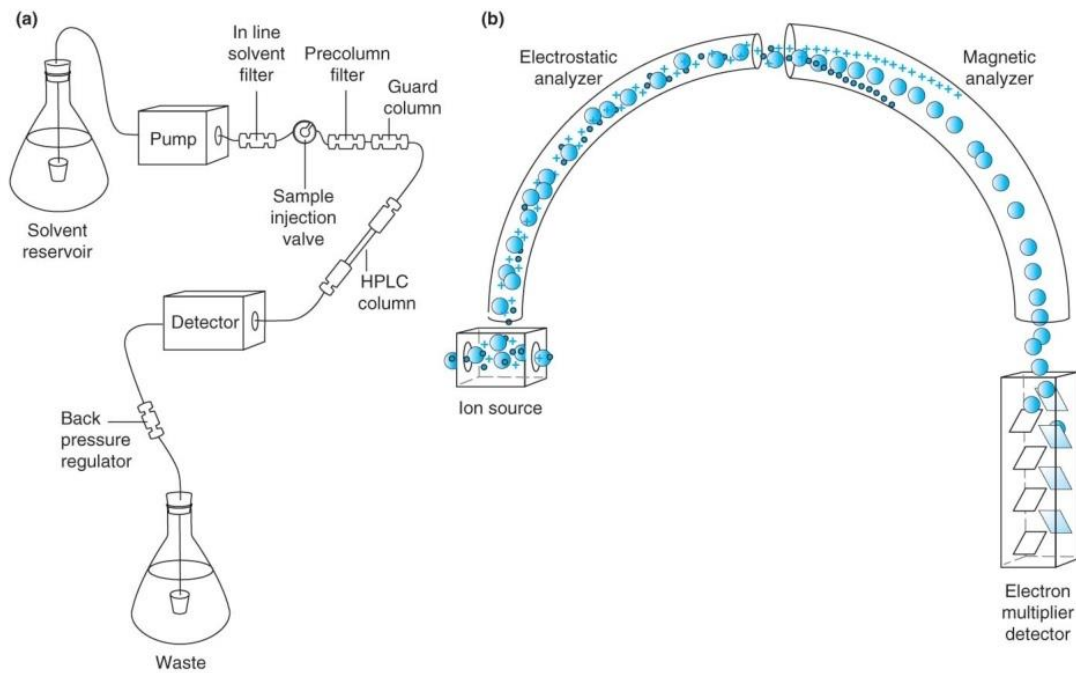
In HPLC we use chromatography column and
push proteins by **pressure**.

We have two types of HPLC

1. Analytical HPLC : can analysis the protein mixture to see how many proteins are there.
2. Preparative HPLC : can used for separate proteins in protein mixture.

Called high performance because quicker and higher resolution.

HPLC contain special type of resin beads smaller and stand the pressure allow better separation.



Mass spectrometry

Is a high sensitive method used for identify small differences between proteins

Used Frequently on the outflow of HPLC systems .

Very important technique for protein purification and analysis in many applications.

It can separate proteins based on mass.

All mass spectrometers do three things :

1. Suspend the mass molecules into the charged gas phase , vaporization and ionization of proteins before analyze by analyzer
2. Separate the molecules based on mass to charge ratio by acceleration down a narrow tube.
3. Detect the separated ions.

Note : after vaporization of proteins and convert to gas phase, the proteins will ionize, all proteins convert to gas and have the same charge.

The analyzer only analyze mass, because all proteins have the same charge.

Analyzer give mass / charge ratio, also we have ion detector.

applications of mass spectrometry

Protein sequencing : Determine of sequence of Amino acids

digest of protein into fragments (peptides) and analyzed to determine the sequence of Amino acids.

Digest of proteins using proteases, before add protein to machine.

M/Z = mass /charge ratio

M - - mass

Z – charge.

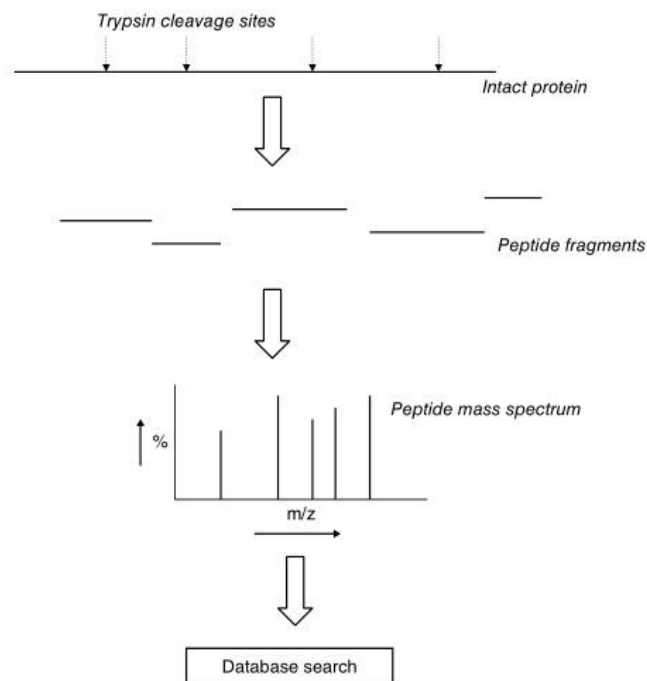
Protein identified : try to determine type and function of protein.

By compare M/Z for every peptide to peptides found in Data base.

Data base :an organized collection of structured information, or Data, typically stored electronically in a computer system.

The major application of M-S is protein sequencing, require two M-S machines in tandem.

MS-based protein identification

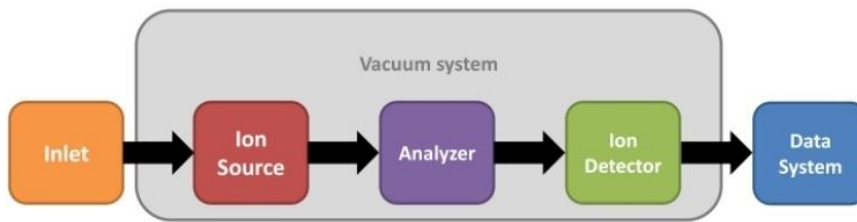


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Component of M-S

The major components inside the machine
(vacuum system)

Analyzer, ion source, and ion detector.



Mass spectrometry used for

1. Determine protein mass
2. Generate partial or full amino acid sequence data for a protein
3. Quantify the amount of protein present in a sample
4. Detect and identify protein PTMs
5. Detect protein modification such as oxidation, deamidation or proteolysis
6. Provide some information on protein structural detail.

Note : many applications of M-S in Proteomic.

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